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NEURAL BELLMAN OPERATORS

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This paper develops Neural Bellman Operators for constructive policy evaluation and feasible improvement. A witness-preserving continuation connects approximation resources to the original economic optimum. Verified comparisons permit trainable proposals without assuming successful unrestricted neural optimization. We prove that a frozen fitted continuation supplies original-law cost observations whose range is controlled by residual oscillation, and construct qualification-optimal fixed-feature readouts with exact primal–dual certificates. Conditional ReLU moments make their evaluation nonrecursive. A predictable cost certificate remains valid under optional stopping and distinguishes fixed-controller costs from adaptive conditional averages. Capacity-repaired witness transport supports paid computational reuse. Matched backward constructions in the nonlinear investment economy qualify actual fitted continuations; the retained scalar and two-control studies separately assess training, deployment and policy-cost equivalence. Complete proofs, economic cost accounts and all prior adverse evidence are retained.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Information acquisition, Computational economics.

1. INTRODUCTION

A numerical continuation matters because it helps produce an economic decision. The error of a fitted value, the loss of the implemented policy and the resources required to verify that loss are different objects. A policy may improve without a tighter numerical certificate; a tighter certificate may leave the policy unchanged. A useful computational method must connect its approximation and resource choices to a feasible decision under the original economic law.

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This paper develops Neural Bellman Operators (NBOs) along that chain. The constructive implementation evaluates a continuation against its own fitted future and retains feasible action witnesses. Its exact affine–ReLU realization and primitive error budget do not require a theorem that unrestricted nonconvex training succeeds. A complementary recursive implementation queries the original Bellman problem only where its action search requires information. A trained readout proposes actions; verified lower and upper bounds determine whether the actual policy meets its original-optimum contract.

The nonlinear investment economy has state-dependent capacity, bilinear transitions and one or two independent continuous innovations. Primitive curvature controls the convexity defect of the transition and supplies regularity of the true Bellman value. The action certificate can therefore dispense with a stored state lattice. A strong-convexity upper chord sometimes certifies a rounded action before another endpoint query. Protected conventional refinement remains available when a proposal fails. The certificate-aware fixed-feature readout and its optimization gap make the resulting training problem constructive. Neither a fitted neural continuation nor an arbitrary proposed action is assumed convex.

Removing a state lattice does not remove the cost of evaluating a future. A first original-value query can generate a large descendant tree. Independently evaluating optimal advantages for precise policy-cost inference can generate another such tree. We address these two costs separately, while retaining the original operator, economic law and optimum.

First, a frozen fitted NBO continuation can replace the optimal value in the cost observation. Its Bellman-residual oscillations and implemented greedy gaps control both the observation's range and the actual policy loss. A terminal correction makes the identity valid even when the fitted terminal value is inexact. State-independent continuation shifts cancel. For shallow affine–ReLU continuations, the original one- and two-innovation conditional expectations are finite positive-part moment calculations. An explicit state-and-action-cover construction qualifies their residuals without querying the optimal Bellman tree. This is an approximation, integration and certification result, not a claim that a fitted critic is already accurate because its training loss is small.

Second, a lower original-Bellman bound and its feasible upper witness can be transported to a nearby state. The rule pays for state displacement, capacity repair and action rounding before deciding whether the old certificate closes at the new tolerance. A failed check invokes the original recursive verifier. The policy guarantee holds after history-dependent selection among valid records, without assuming equality with the fresh-search policy. A finite-cover theorem states the tighter fill precision and retention conditions needed to bound the number of expensive fills. The executed bounded cache is evaluated by its actual counts, rather than by premises it does not satisfy.

These additions distinguish three uses of a neural object: a constructed continuation, an optional query proposal and a proposed record for retrieval. Their economic value is not identified by calling a common correctness argument neural. The neural continuation now enters a computable original-law evaluation identity; the proposal and retrieval mechanisms can change work but cannot manufacture a value bound. Exact ReLU encodings and conventional piecewise-linear interpretations remain related representations, not independent algorithms with an automatic ranking.

The retained frozen study executes certificate-aware ReLU and equally sized Bernstein readouts, independent-context validation, sixty-four-block reuse, direct continuous-law policy-cost inference and trained two-control comparisons. Its twelve scalar cost contrasts meet the prespecified equivalence margin, but none identifies strict superiority. Its complete cold-start evidence does not support a uniform neural work advantage. The new predeclared experiment compares fresh calculation, exact-key reuse and geometrically transported witnesses on the same localized query workloads, including a four-date, two-control stress case. It counts every original-law descendant on a miss and every rejected lookup, invalidation and durable record. The workload is not relabeled as independent policy-cost data.

Fitting for the economic qualification. The continuation's attained fitting criterion should control a quantity that enters the policy decision. Section 9 minimizes a verified residual-oscillation bound over a fixed feature class, conditional on its actual fitted future. A finite linear program proposes the readout; an exact primal–

dual calculation certifies its achieved objective gap. Backward construction links those attained bounds to original-optimum regret and to the range of the original-law cost observation. The matched ReLU and conventional constructions below execute this chain, including continuous-action coverage and whole-cell feasibility. They do not replace a verified uniform bound by a sampled training loss.

A second decision concerns when enough qualification has been purchased. Section 10 permits predictable continuations and range qualifications and remains valid when a cost comparison is stopped after inspection. It distinguishes repeated qualification of a fixed complete controller from the conditional average cost of changing controllers. Failed numerical evaluations consume their prequalified support and actual work instead of disappearing from the sample. The adoption account then prices qualification and replacement without treating processor seconds as money or the sample count as the deployment horizon.

Relation to existing methods. Lipschitz extension and distance transforms underlie the constructive envelope (McShane, 1934, Felzenszwalb and Huttenlocher, 2012). Approximate and safe policy iteration have established error and performance analyses (Munos, 2003, Pirotta et al., 2013, Scherrer et al., 2015); Yang (2020) studies convex dynamic-programming approximation. Learned warm starts and optimization-based control have their own literature (Sakaue and Oki, 2023, Sambharya, Hall, Amos, and Stellato, 2023, Agrawal, Barratt, Boyd, and Stellato, 2020). Our contribution connects those numerical choices to an acquired feasible policy, a verified stopping decision and the complete cost of returning and qualifying the controller.

Value-function control variates and telescoping identities are established evaluation tools (Munos, 2006, Jiang and Li, 2016). Here a verified residual-oscillation bound supplies both an original-optimum policy account and the concentration range of the actual original-law cost difference. Exact conditional ReLU integration and the explicit residual-cover account state what must be purchased to replace optimal-value centering by a fitted continuation. The witness-transfer theorem likewise supplements ordinary memoization with a capacity-feasible upper witness and a lower bound on the original continuous-action infimum.

Time-uniform concentration through nonnegative supermartingales is established (Howard, Ramdas, McAuliffe, and Sekhon, 2020); off-policy confidence sequences have been developed for contextual-bandit evaluation (Karampatziakis, Mineiro, and Ramdas, 2021). We use those principles rather than claim them as new. Our observations are directly simulated original-law NBO contrasts, enclosed over continuous innovation cells and bounded by a qualified fitted continuation. The readout dual and the predictable support construction identify the numerical and economic objects to which optional stopping applies.

Reading path. Sections 2–5 give the original constructive operator, investment primitives and recursive guarantee. Sections 6–7 state the operational readout and two-control certificates. Sections 8–10 develop fitted-continuation evaluation, qualification-optimal readouts and sequential cost certification. Section 11 retains paid-witness transport and Section 12 reports the evidence. The Technical Supplement includes optimal-advantage centering and all complete proofs. The Numerical and Development Appendix retains the preceding economic, localization, deployment and empirical sections unchanged, and adds the detailed qualification records. Original broader applications and the preceding complete editions remain companions under their stated hypotheses.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t, the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j<t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v, define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \qquad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \tag{1}$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous

objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. The direct-cost inference in Section A is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1}, returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. *Action-dependent errors and complete-policy loss*

Let $H(a)$ be a true current cost and $\widehat{H}(a)$ its continuation approximation. If $\widehat{a}$ is δ-optimal for $\widehat{H}$ on the same feasible action set, then

$$H(\widehat{a}) - \inf_a H(a) \le \delta + \operatorname{osc}_a\{\widehat{H}(a) - H(a)\}. \tag{2}$$

Indeed, compare $\widehat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \le u_t, \qquad \mathcal{T}_t^{\pi} f_{t+1} - f_t \le v_t, \qquad \ell_T \le f_T - g \le u_T.$$

Then the actual feasible policy satisfies

$$0 \le J_0^\pi - V_0^* \le \sum_{t<T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \tag{3}$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

The proof is in Supplement, Section 2.1.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \le K_t^A \|x - z\|_1, \qquad L_t = A_t^Q + D_t K_t^A, \qquad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t-Lipschitz whenever f_{t+1} is L_{t+1}-Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t, form effective feasible k_t-nets. Evaluate their own-future objectives with certified error at most e_t. Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \le K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \tag{4}$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg\min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \tag{5}$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g.

THEOREM 2—Own-future construction with feasible witnesses: *The functions in* (5) *have exact affine–ReLU realizations and Lipschitz moduli* L_t *independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \le f_t - \mathcal{T}_t f_{t+1} \le D_t k_t + e_t + 2L_t h_t, \tag{6}$$
$$\mathcal{T}_t^{\pi} f_{t+1} - f_t \le e_t + D_t \zeta_t,$$

and hence

$$J_0^{\pi} - V_0^{*} \le \sum_{t<T} B_t(D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T(2e_T + 2L_g h_T). \tag{7}$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

The proof is in Supplement, Section 2.2.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

3.1. *Original-owner compilation on a tensor cover*

On a tensor cover $\mathcal{X} = \prod_{j=1}^{d} \{x_{j,0}, \dots, x_{j,n_j}\}$, put $S = \prod_j (n_j + 1)$. For arbitrary labels y_i, including inconsistent inexact labels, compute at each grid node v

$$(z_v, o_v) = \min_{\text{lex,i}} (y_i + L\|v - x_i\|_1, i).$$

The index o_v is the original label owner, not the transformed node v.

THEOREM 3—Continuous original-owner identity: *For a closed tensor cell C with corners $\mathcal{V}(C)$, and $x \in C$,*

$$f(x) = \min_{v \in \mathcal{V}(C)} (z_v + L\|x - v\|_1), \tag{8}$$

$$(f(x), i_*(x)) = \min_{\text{lex,v} \in \mathcal{V}(\text{C})} (y_{o_v} + L\|x - x_{o_v}\|_1, o_v).$$

Coordinatewise forward and backward relaxations compute (z, o) in $O(dS)$ additions and comparisons and $O(S)$ storage. After locating the cell, a query uses $O(d2^d)$ arithmetic operations. Exact rational comparisons preserve all ties, including $L = 0$. Using the original owner's repaired action preserves the uncompiled policy, its cost and its certificate exactly.

The proof is in Supplement, Section 2.3.

The nodal algorithm is the classical distance transform (Felzenszwalb and Huttenlocher, 2012); preserving the original action through continuous off-grid queries is the policy obligation. If there are $J_{t,i}$ node actions and M_t integration representatives, write $Q_t = M_t \sum_i J_{t,i}$. Literal dense cone evaluation costs $O(\sum_t dS_{t+1}Q_t)$ in representation work. Compilation replaces this by $O(\sum_t [dS_{t+1} + Q_t d2^d])$, plus cell location, primitive evaluation, integration, feasible repair, certificate and output. Rational bit lengths and state boxes crossing many cells incur additional costs. There is no dimension-free complexity assertion.

3.2. *Acquired states and conventional actors*

A b-bit coordinate sensor reports a dyadic cell and its midpoint $\widehat{x}$. On the unit cube, $\|x - \widehat{x}\|_1 \le r(b) = d2^{-b-1}$. For the investment capacity $\kappa(x)$ below, clip the original action to the smallest capacity throughout that cell and round down to an action quantum Δ_t. This is feasible at every state in the cell. Relative to an exact true-state repair, its displacement is at most $2K_t^A r(b) + \Delta_t$. Transporting the witness cone from $\widehat{x}$ to x adds at most $2L_t r(b)$, plus any certified selector objective error ν_t. Thus the implemented-policy account adds

$$\sum_{t<T} B_t\{(2L_t + 2D_t K_t^A)r(b) + D_t\Delta_t + \nu_t\}. \tag{9}$$

No continuity of the discrete owner selector is assumed. At uncertain numerical comparisons the simulator retains all possible actors, rather than choosing a favorable one.

For conventional FVI the continuation is multilinear on its realized tensor grid. Its actual modulus is bounded by the largest coordinate edge slope. A nearest-node actor uses a feasible node minimizer and true-state repair. Let h_t be the largest half-cell ℓ^1 radius, L_t^{gr} the Bellman graph modulus, and e_t the node query allowance. Its optimal residual upper bound is $D_t k_t + e_t + L_t^{\mathrm{gr}} h_t$. Its selected-policy upper bound is $e_t + E_t$, where

$$E_t = \sup_x \{ y_{i(x)} + L_t^{\mathrm{gr}} \| x - x_{i(x)} \|_1 - f_t(x) \}.$$

Partitioning cells by nearest-node boundaries makes the expression separately affine in each coordinate. Its maximum is therefore at a partition vertex; the implementation encloses all such vertices. Both uniform and curvature-refined FVI receive this sharper one-sided account, not an artificially doubled baseline. Their acquired-state allowance is derived separately in (6).

4. ORIGINAL-OPTIMUM ACCURACY IN THE INVESTMENT ECONOMY

The exact witness theorem answers an implementation question conditional on a continuation. An economist also needs to know how close the complete implemented policy is to the original optimum. We now establish that connection in the investment economy itself. The argument exploits a property not used by the earlier cell-range calculation: despite the bilinear transition, the Bellman objective remains jointly convex. This permits a second-order interpolation account and transfers verified nodal decisions to an implementable policy over the entire state domain. It does not assume that a fitted neural continuation is convex.

Relation to the preceding two-control query. The earlier query experiment used the fixed shared capacity $1/8$. The complete dynamic extension below uses the original state-dependent capacity $c(x)$, with the retained exposure columns, action-cost interaction and second shock. Its nesting claim concerns the original scalar

economic model; it is not an assertion that a full dynamic policy was already present in the earlier isolated query.

4.1. *Primitives and Bellman regularity*

Let $d \geq 2$ be even, let indices be cyclic, and set $\beta = 15/16$. Write

$$S(x) = \frac{2}{d}\sum_j (x_j - 5/8)^2 + \frac{1}{4d}\sum_j (x_j - x_{j+1})^2 + 2\left(\frac{1}{2} - \frac{2}{d}\sum_j x_j\right)_+^2,$$

$$g(x) = S(x) + \frac{2}{d}\sum_j (x_j - 5/8)^2, \qquad c(x) = \frac{1}{8} + \frac{1}{8d}\sum_j x_j.$$

The original scalar control has $0 \leq a \leq c(x)$, running cost $S(x) + a^2 + 4a^4$, and transition

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + B_j a + s_j z,$$

where $(B_j, s_j) = (1/2, 1)$ at alternating coordinates and $(1/4, -1)$ at the others, and z is uniform on $[-1/32, 1/32]$.

For the retained two-control extension, $a_1, a_2 \geq 0$, $a_1 + a_2 \leq c(x)$, the second exposure column interchanges $1/2$ and $1/4$, and the action cost is

$$R(a) = a_1^2 + a_2^2 + 4(a_1^4 + a_2^4) + a_1 a_2/4.$$

An independent common-direction uniform innovation of radius $1/64$ is added. This is the previously specified extension, nested at $a_2 = 0$ when that innovation is disabled. Both models preserve $[0,1]^d$. Let $m \in \{1, 2\}$ denote the number of controls and define

$$Q_t^*(x, a) = S(x) + R(a) + \beta \mathbb{E} V_{t+1}^*(F(x, a, Z)),$$

$$V_t^*(x) = \min_{a \in A(x)} Q_t^*(x, a), \qquad V_T^* = g.$$

All costs and laws in this display are the economic primitives, not a fitted surrogate.

THEOREM 4—Regularity of the original Bellman problem: *For either control specification and every finite horizon, V_t^* is convex and continuously differentiable in the interior,*

with the corresponding one-sided extension on the cube. It is G_t-Lipschitz in the ℓ^1 norm and M_t-semiconcave in the Euclidean norm, where valid constants are

$$G_T = 10/d, \qquad G_t = \frac{1037}{128d} + \beta\frac{47}{64}G_{t+1},$$
$$M_T = 26/d, \qquad M_t = \frac{22}{d} + \frac{21}{256d} + \beta\left(\frac{9}{16}M_{t+1} + \frac{1}{8}G_{t+1}\right).$$

In particular, $G_t \le 27/d$ and $M_t \le 54/d$. The function Q_t^* is jointly strongly convex on its convex feasible graph, with state curvature at least $107/(128d)$ and action curvature at least $7/4$.

A valid semiconcavity constant along each coordinate line is

$$D_T = \frac{9}{d} + \frac{16}{d^2},$$
$$D_t = \frac{5}{d} + \frac{16}{d^2} + \frac{21}{256d^2} + \beta M_{t+1}\left(\frac{85}{256} + \frac{23}{256d}\right).$$

Consequently, for the positive multilinear interpolant I_n on a mesh of spacing $1/n$,

$$0 \le I_n V_t^*(x) - V_t^*(x) \le \kappa_t, \qquad \kappa_t = \frac{dD_t}{8n^2}. \tag{10}$$

The proof is in Supplement, Section 2.4.

The theorem is a structural statement about these economic primitives. It is not a claim that every controlled diffusion, recursive preference specification, or game in the complete development edition has a convex Bellman problem. Nor does it require a signed-weight neural fit to preserve convexity. Convex approximation of dynamic programs has an established literature; the additional work here is to verify the needed curvature for the original nonlinear capacity-constrained model and connect it to the actual NBO actor and complete error account.

4.2. *Certified nodal Bellman bounds*

Suppose nodal arrays satisfy $l_{t+1} \le V_{t+1}^* \le u_{t+1}$ at all grid vertices. Equation (10) gives the functions $I_n l_{t+1} - \kappa_{t+1}$ and $I_n u_{t+1}$ as valid lower and upper continuations. The lower function need not itself be convex; convexity is used for the unknown

true value and for the comparison argument, not asserted of arbitrary endpoint arrays.

Divide each uniform innovation component into q equal bins and use its conditional mean. Convexity gives a lower quadrature bound for the true continuation. Semiconcavity gives an upper error

$$\zeta_{t+1} = \frac{M_{t+1}}{6q^2} \sum_{r=1}^{s} \rho_r^2 \|s_r\|_2^2, \tag{11}$$

where $s = 1$ or 2, $\rho_1 = 1/32$, and $\rho_2 = 1/64$ in the extension. The expectation remains the original continuous law. At the terminal date g is evaluated directly, so no terminal interpolation allowance is needed.

At a vertex v, let $\mathcal{A}_A(v)$ be the capacity simplex lattice with spacing $c(v)/A$, including all its faces. For $m = 1, 2$, a minimizer can be represented by neighboring lattice vertices whose coordinate variances sum to at most $mc(v)^2/(4A^2)$. A valid action smoothness constant is

$$H_t = \frac{21}{4} + \beta M_{t+1} \|B\|_2^2, \quad \|B\|_2^2 = \frac{5d}{32}\ (m = 1), \quad \|B\|_2^2 = \frac{9d}{32}\ (m = 2).$$

Taylor's inequality and that barycentric representation imply

$$0 \le \min_{a \in \mathcal{A}_A(v)} Q_t^*(v, a) - V_t^*(v) \le \delta_t := \frac{mH_t}{128A^2}. \tag{12}$$

The linear Taylor terms cancel even when the optimum lies on a face. This is an error against the continuous feasible optimum, not only against the finite action menu.

Let $q_t^-(v, a)$ and $q_t^+(v, a)$ include all interpolation, quadrature and arithmetic errors and enclose $Q_t^*(v, a)$. Then

$$l_t(v) = \max\{0, \min_{a \in \mathcal{A}_A(v)} q_t^-(v, a) - \delta_t\}, \qquad u_t(v) = \min_{a \in \mathcal{A}_A(v)} q_t^+(v, a) \tag{13}$$

are valid nodal Bellman bounds. The zero truncation uses the nonnegative economic costs. Extra neural or conventional proposals may improve the upper side but do not remove the lower action-cover allowance. Positive interpolation, averaging and minimum maps are monotone and nonexpansive; the separate allowances in

(10)–(12) are therefore propagated, rather than being discarded because an endpoint calculation contains the truth.

4.3. *From a fitted witness to the complete implemented policy*

Let $a_t(v) \in A(v)$ be the actual action returned at each vertex by any generator. It may be selected by the exact neural witness, a conventional fitted continuation, or their common verified fallback. Write $w_v(x)$ for the multilinear weights and set $\bar{\pi}_t(x) = \sum_v w_v(x) a_t(v)$. Affineness of capacity makes this action feasible. Let $q_t^{\mathrm{sel},+}(v)$ be a verified upper bound on its original-optimum continuation value $Q_t^*(v, a_t(v))$, using the same lower and upper Bellman reference for all generators.

THEOREM 5—Actual-policy, whole-domain certificate: *Set*

$$e_t = \max_v \{ q_t^{\mathrm{sel},+}(v) - l_t(v) \} + \kappa_t.$$

Suppose the implemented policy $\widehat{\pi}_t$ is feasible and differs from $\bar{\pi}_t$ by at most b_t in action ℓ^1 norm, uniformly over the state cube. Define

$$\Lambda_t = \frac{13}{16} + \beta \frac{3d}{8} G_{t+1}, \qquad D_T^\pi = 0, \quad D_t^\pi = e_t + \Lambda_t b_t + \beta D_{t+1}^\pi.$$

Then, at every true state and date,

$$0 \le J_t^{\widehat{\pi}}(x) - V_t^*(x) \le D_t^\pi.$$

This statement is conditional on the actual recorded policy, not on success of its fitting algorithm. It applies equally to a pure fitted policy and to a policy augmented by verified common candidates.

The proof is in Supplement, Section 2.5.

In the execution, observation is rounded down at forty bits. An acquired state $y \le x$ has no greater capacity than the true state. Each interpolated action is then rounded down at twenty bits using the lower endpoint of its arithmetic enclosure. If L_{ij} bounds the derivative of action coordinate i with respect to state coordinate j,

and η bounds native interpolation error, a valid implementation allowance is

$$b_t \le 2^{-40} \sum_{i,j} L_{ij} + m(2^{-20} + 2\eta + \eta_{\mathrm{out}}).$$

Here L_{ij} is computed from the maximum adjacent nodal difference times n, and η_{out} covers endpoint subtraction. All these numbers are recorded outward. This new barycentric acquired actor is explicitly distinguished from the earlier discontinuous cell-constant actor; an old actor is not silently replaced during a purported fixed-policy revalidation.

4.4. *Arithmetic, work, and scope*

The native interpolation kernel forms nonnegative tensor weights and a weighted sum of at most 2^d stored binary64 endpoints. It is compiled without reassociation or floating-point contraction. For $K = (3d + 4)2^d + 32$, an explicit conservative absolute allowance is

$$\eta = \frac{K2^{-53}}{1 - K2^{-53}} \max_v |z_v| + K2^{-1022}.$$

The final term conservatively covers underflow. Input coordinates, affine indexing, finite endpoints and the binary64 format are checked. Roundoff expansion and the ordinary interval cost arithmetic remain separate from the analytic approximation bounds. Exact rational fixtures compare the native sum with its mathematical tensor interpolant.

With $N = (n + 1)^d$ vertices, $K_A = A + 1$ scalar actions or $K_A = (A + 1)(A + 2)/2$ two-control actions, the reference calculation uses $O(TNK_A q^s 2^d d)$ elementary work under the explicit tensor implementation. Stored reference values use $O(TN)$ words and a policy uses $O(TNm)$ words. Streaming batches avoid a state-pair table. Proposal fitting, exact rational witness work, actor transfer, serialization, independent cost inference and compilation are additional costs, not absorbed into this bound. The analytic errors are second order in state, action and innovation resolution; the state-count and tensor-evaluation factors still depend exponentially on dimension. The result supplies a matched original-optimum comparison without asserting that these costs have disappeared.

5. NEURAL BELLMAN QUERIES WITHOUT A STATE LATTICE

The preceding construction makes numerical error accountable, but a complete tensor reference can absorb the benefit of a fitted continuation. We therefore change the computational organization, not the economic problem. A Neural Bellman selector proposes an action at the state actually acquired. A recursive interval query then bounds the original Bellman value at that state and checks the proposed action against the entire continuous feasible interval. No conventional policy is precomputed at all states. The controller is the resulting callable algorithm, not the unverified proposal alone.

The distinction matters for attribution. Learning may reduce the work needed to locate a useful action. It does not supply the inequalities used to certify that action. The adaptive conventional comparator receives the same inequalities and the same original-optimum target. The trained value-fit policies in the preceding study remain distinct from the action-query predictor introduced here.

5.1. *A curvature lower bound on an adaptive action partition*

Use the original scalar economy of Section 4, with r dates remaining. Write V_r for its optimal value and $Q_r(x,a)$ for the value of a first action followed by the optimal future. Let G_r, M_r be the primitive constants in Theorem 4, indexed by remaining horizon, and put

$$H_r = \frac{21}{4} + \beta M_{r-1}\frac{5d}{32}, \qquad \Lambda_r = \frac{13}{16} + \beta G_{r-1}\frac{3d}{8}.$$

Thus H_r is an upper curvature bound in the scalar action, and Λ_r bounds the absolute action derivative. Neither depends on a fitted model.

LEMMA 6—Parabolic minorants: *Suppose f is H-semiconcave on $[l,u]$, and $L_l \le f(l)$, $L_u \le f(u)$. For $s \in [0,1]$,*

$$f(l + s(u-l)) \ge (1-s)L_l + sL_u - \frac{H}{2}(u-l)^2 s(1-s) =: p(s).$$

Consequently $\min_{[0,1]} p$ is a lower bound for the minimum of f on the interval. With $K = H(u-l)^2/2$ and $D = L_u - L_l$, this lower bound is L_l if $D \ge K$, L_u if $D \le -K$, and

otherwise

$$\frac{L_l + L_u}{2} - \frac{K}{4} - \frac{D^2}{4K}.$$

The endpoints L_l, L_u need not themselves belong to a convex interpolant. Increasing K preserves the lower-bound property.

The proof is in Supplement, Section 2.6.

At an acquired state, retain an ordered partition of the feasible action interval and an outward value bracket at every partition point. The minimum of the parabolic minorants bounds the continuous-action infimum from below. The smallest upper endpoint supplies an actual feasible witness. A neural or polynomial proposal inserts another point into this partition; it cannot remove the endpoints or any part of the feasible interval. The running lower bound is the maximum of all previously valid global lower bounds, so new information cannot weaken the certificate. The upper bound is the minimum over every evaluated feasible witness.

5.2. *Recursive queries under the original continuous law*

For an endpoint action with $r > 1$, divide the scalar innovation into q_r equiprobable bins and query V_{r-1} at each conditional-mean next state. If each child returns width at most η_r, the endpoint bracket has width at most

$$\beta\eta_r + \frac{\beta M_{r-1} d}{6144 q_r^2} + \omega_r. \tag{14}$$

Here ω_r includes outward primitive arithmetic and the Lipschitz expansion from an interval-valued next state to its queried center. Convexity provides the lower conditional-mean inequality; semiconcavity and the exact conditional variance provide the upper allowance. The innovation remains continuous. There is no state-interpolation term because a child value is queried rather than read from a state lattice.

For a requested local gap δ, the implementation sets $\eta_r \le \delta/(4\beta)$ and chooses a dyadic q_r so that the second term in (14) is at most $\delta/4$. A separate endpoint-width guard charges arithmetic and the small capacity-rounding remainder. Failure of

that guard is a precision failure, not a successful return. The terminal query uses analytic integration instead of recursive quadrature.

THEOREM 7—A state-grid-free NBO service: *Consider the original scalar investment economy. At every recursive query, suppose the outward endpoint and capacity-transfer accounts give width at most $5\delta/8$. Let the lower bound be the running maximum described above. Refine an interval attaining the smallest current minorant whenever the global upper-minus-lower gap exceeds δ. Each new point must lie between one fifth and four fifths of that interval. Arbitrary additional learned proposals may be inserted initially. Then:*

(i) every returned interval $[L_r(x), U_r(x)]$ contains $V_r(x)$, and its returned feasible action $a_r(x)$ satisfies

$$Q_r(x, a_r(x)) - V_r(x) \le U_r(x) - L_r(x) \le \delta;$$

(ii) the real-arithmetic refinement terminates. With at most one interior learned proposal, a sufficient point budget is

$$6 + \left\lceil 5c(x)\sqrt{H_r/(3\delta)} \right\rceil; \tag{15}$$

(iii) for the policy that calls this service at every subsequent date, the original-law loss satisfies

$$J_T^{\pi}(x) - V_T(x) \le \sum_{t=0}^{T-1} \beta^t \left[\delta_t + \left(8 + \frac{11\beta d G_{T-t-1}}{16} + d G_{T-t} \right) 2^{-b} \right], \tag{16}$$

when the state is acquired downward at b bits per coordinate and every returned action is feasible at that acquired state. All action rounding must precede its endpoint evaluation. The inequalities hold uniformly in the true initial state, not only at the states used in an experiment.

The proof is in Supplement, Section 2.7.

The finite-precision contract is explicit. The executed service uses the width guard, a 512-point cap, and a 32-bit downward action lattice. A precision or cap failure withholds the floating certificate and invokes a separately counted exact-rational enumeration. This fallback constructs values only at the required descendants; it does not load a precomputed conventional policy. The rational fallback chooses

dyadic action and quadrature counts until each cover or quadrature allowance is at most one quarter of the request, and requests child width at most one quarter divided by the discount. Provided $\delta > 8\Lambda_r 2^{-32}$ at every recursive level, its exact brackets have width at most three quarters of the request. This proves a finite return for every acquired state in the declared tasks, including states absent from the workload. Rational bit-operation cost is additional work and is not replaced by a unit-cost floating clock. The mathematical point budget is checked against the declared tolerance range; termination in an arbitrary floating implementation is not inferred merely from the real-arithmetic argument. Fitting convergence is unnecessary for validity: any finite prediction, and even its clipped replacement when numerically invalid, is only a proposed query.

5.3. *A certified analytic terminal kernel*

The last decision admits an especially strong common comparator. Let $y = F(x,0,0)$, $b_0 = 1/2 - 2\,\mathrm{mean}(y)$, and $\gamma_j = B_j$. Define

$$D(x) = \frac{8}{d}\sum_j \gamma_j (y_j - 5/8) + \frac{1}{2d}\sum_j (\gamma_j - \gamma_{j+1})(y_j - y_{j+1}).$$

For every even d, the exact terminal action derivative is

$$16a^3 + \left(2 + \frac{41\beta}{32}\right) a + \beta D(x) - 3\beta(b_0 - 3a/4)_+. \tag{17}$$

Each branch is a strictly increasing cubic. Its floating closed-form root supplies only a candidate. The independently evaluated outward derivative and exact-law objective certify that candidate; a safeguarded derivative bracket remains available if the candidate is inaccurate.

For completeness, if f is μ-strongly convex and an outward interval encloses $f'(a)$, the largest compatible projected derivative residual R gives

$$f(a) - \min_{[0,c]} f \le R^2/(2\mu).$$

At zero, a nonnegative derivative has residual zero; at capacity, a nonpositive derivative has residual zero. Otherwise the absolute derivative bound is sufficient.

This follows by minimizing the strong-convexity supporting quadratic over the feasible interval and, conservatively, over the line. Here $\mu = 2$. Equation (17), its $41/32$ and $9/4$ coefficients, analytic shock moments, and outward derivatives are independently checked against rational formulas. Every method uses this kernel; it is not credited to the neural predictor.

5.4. *Work, prediction, and the economic object*

Let $A_r(\delta)$ be a bound on action endpoints at a query. Excluding training, a scalar operation account obeys

$$W_1(\delta) = O(dA_1), \qquad W_r(\delta) \le CdA_r(\delta) + A_r(\delta)q_r W_{r-1}(\delta/(4\beta)).$$

The constants dG_r, dM_r are bounded independently of d, so the declared endpoint and quadrature budgets need not grow exponentially with the state dimension. Primitive evaluations still cost $O(d)$. This removes the state lattice, not the horizon or shock-tree cost. The executed vectorization batches a query's innovation descendants. Its largest live state batch is reported; it does not claim the smaller memory of a fully streamed implementation. Increasing the horizon or the shock dimension may still be expensive. The new execution is scalar-control; the earlier full two-control result is retained rather than relabeled as a query-service experiment.

A useful prediction can reduce the endpoint count or return its own action from the initial partition. An inaccurate prediction can instead add work. This is a prediction-dependent algorithm, not a theorem of neural dominance. Learned warm starts have their own literature, including Sakaue and Oki (2023); their discrete-convex setting differs from the original-law Bellman intervals here. Convex control policies and learned optimization parameters are studied by Agrawal, Barratt, Boyd, and Stellato (2020). Our comparison identifies the extra proposal cost and the resulting query transcript rather than claiming priority for learning a warm start.

The purchase is now a complete callable controller with a declared error contract. Its training cost, every subsequent certified query, and its return-time cost belong to the same service. A reduction in queried state storage is economically relevant for repeated decisions only after these costs are counted. The original cost normalization

is maintained. Workload seconds and normalized expected-cost loss are reported separately; a price per unit of computation can convert the former to the latter only as an explicitly specified decision-maker parameter.

6. OPERATIONAL PREDICTION CERTIFICATES AND READOUT TRAINING

Proposition 5 separates witness accuracy from lower-certificate localization. Its distance to an unknown minimizer is useful for analysis but is not a stopping observable. We now replace that distance by an upper envelope computed from information already purchased. This gives a certificate before an additional Bellman query, a training criterion measured in unresolved verification work, and a constructive neural readout with a computable optimization gap.

6.1. *A decision before the next query*

Fix a state, a date, and the current action partition. Write $f(a) = Q_t^*(x,a)$, and suppose f is μ-strongly convex on the feasible interval. Let $\overline{f}_l, \overline{f}_u$ be verified upper endpoints at two consecutive actions $l < u$. Let L be the current lower bound for the original feasible infimum, including the capacity and arithmetic allowances. For $a = l + s(u-l)$, define

$$C(a) = (1-s)\overline{f}_l + s\overline{f}_u - \frac{\mu}{2}(a-l)(u-a), \qquad 0 \le s \le 1. \tag{18}$$

Every symbol in this expression is either a primitive bound or an existing endpoint. No derivative or action at the unknown optimum occurs.

THEOREM 8—Pre-query closure: *Under the preceding conditions, $f(a) \le C(a)$ throughout $[l,u]$. Let $a_\Delta \in [l,u]$ be the actual feasible rounded proposal. An outward upper endpoint $\overline{C}(a_\Delta)$ satisfying*

$$\overline{C}(a_\Delta) - L \le \delta \tag{19}$$

certifies $f(a_\Delta) - \inf_{a \in A(x)} f(a) \le \delta$ without another Bellman query. If the screen does not close, the protected refinement of Theorem 4 retains its validity and finite query bound. The complete acquired policy therefore retains the original-optimum account of Theorem 7.

Equivalently, the union over partition intervals of the sublevel sets $\{a : C(a) - L \le \delta\}$ is a computable certified action set. Each constituent is a quadratic sublevel interval intersected with its partition cell. Endpoints obtained numerically must be rounded inward, or the final action must be checked by (19).

Proof. See Supplement, Section 3.1.

The screen is shared by every comparator. In particular, minimizing the quadratic upper envelope is an available conventional proposal. Prediction is valuable only to the extent that its proposed location or evaluation cost improves the observed service. The executed neural rule screens its feasible prediction on the partition interval containing it, even when that location is outside the protected part of the currently critical interval. A failed screen does not authorize an unprotected split.

6.2. *A computable loss for unresolved work*

A *certificate context* z consists of the state, remaining horizon, tolerance, partition endpoints and their verified bounds. A proposal $a_\theta(z)$ is feasible and is computed without another Bellman query. Define its verified score and a margin loss by

$$s_\theta(z) = \overline{C}_z(a_\theta(z)) - L_z, \tag{20}$$

$$\ell_\tau(z,\theta) = \min\left\{1, \frac{(s_\theta(z) - \delta_z + \tau)_+^2}{\tau^2}\right\}, \qquad \tau > 0. \tag{21}$$

Let $K(z)$ be a valid cap on further endpoint queries if the screen fails and the protected fallback is followed. This is a count of endpoint queries at the current state. The recursive cost of each endpoint is charged through the earlier descendant-work recursion. On contexts satisfying the declared precision contract, the real-arithmetic and rational-recovery bounds supply finite such caps.

THEOREM 9—Certificate loss and additional work: *If $N_\theta(z)$ is the number of additional endpoint queries after the screen, then*

$$N_\theta(z) \le K(z)\mathbf{1}\{s_\theta(z) > \delta_z\} \le K(z)\ell_\tau(z,\theta). \tag{22}$$

Consequently, an achieved margin loss bounds the average additional queries on any specified finite context catalogue, with no unknown minimizing action.

For a distributional assertion, condition on construction of a finite family of M predictors. Suppose $z_1, \dots, z_n$ are independent validation contexts from a declared predictor-independent law, and $0 \le K(z) \le K_{\max}$. With probability at least $1 - \alpha$, simultaneously for that family,

$$\mathbb{E}N_\theta(z) \le \frac{1}{n}\sum_{i=1}^{n} K(z_i)\ell_\tau(z_i,\theta) + K_{\max}\sqrt{\frac{\log(M/\alpha)}{2n}}. \tag{23}$$

The cost of obtaining and validating contexts and of evaluating the predictor is additional work, not part of N_θ.

Proof. See Supplement, Section 3.2.

The context-law requirement matters. A stream of recursively generated contexts can depend on earlier predictions; it is not automatically an independent validation sample for (23). The deterministic inequality (22) remains valid context by context. The recorded action-label mean squared error is not substituted for (21). The new loss measures a certificate's unresolved work rather than resemblance to an approximate action label.

6.3. *A constructive ReLU readout with an optimization certificate*

The work loss can be trained without invoking a successful hidden-layer optimizer. Fix a nonnegative affine–ReLU feature map $\phi(z) \in \mathbb{R}^p$ with $\sum_j \phi_j(z) = 1$. Linear hat functions on a fixed coordinate partition are one exact construction; averages of such maps give multivariate features. A box-constrained readout $w \in [0,1]^p$ proposes

$$a_w(z) = l_z + (u_z - l_z)\phi(z)^\top w.$$

The partition interval and features are fixed before optimizing w. Let $\eta_i \ge 0$ cover action rounding and evaluation of the score, so that the computed score of the actual rounded action is at most $C_i(a_w(z_i)) + \eta_i - L_i$. For example, on lattice-aligned endpoints an action Lipschitz bound gives $\eta_i = \Lambda_i \Delta_a$ for downward rounding.

Define

$$F(w) = \frac{1}{n}\sum_i K_i\big(C_i(a_w(z_i)) + \eta_i - L_i - \delta_i + \tau\big)_+^2 + \frac{\lambda}{2}\|w\|_2^2, \qquad \lambda \geq 0. \tag{24}$$

Unlike an action-label objective, this loss directly upper-bounds the additional-work account. It trains the upper witness against the current global lower certificate.

PROPOSITION 10—Certified readout training: *The function F is convex and continuously differentiable on the box. For $g = \nabla F(w)$, put*

$$D(w) = g^\top w - \sum_{j=1}^{p} \min\{0, g_j\}. \tag{25}$$

Then $0 \leq F(w) - \min_{v\in[0,1]^p} F(v) \leq D(w)$. On the training contexts the average additional endpoint-query count is at most $F(w)/\tau^2$. If $\lambda > 0$, the minimizer w^* is unique and

$$\|w - w^*\|_2 \leq \sqrt{2D(w)/\lambda}.$$

All quantities in these conclusions can be bounded by outward arithmetic; they are exact rational quantities when the contexts, features and iterate are rational.

Proof. See Supplement, Section 3.3.

This is a fixed-feature neural construction, not a convergence assertion for unrestricted nonconvex network fitting. The deposited rational implementation uses finite conditional-gradient steps and reports its achieved loss and dual gap. Its regression examples test the proposition; they are not new investment-policy observations. The preceding R67 service comparison retained its preregistered action-label fits. The new comparison in Section F separately executes this certificate-aware objective, with equal-information Bernstein and label-loss controls; no historical fit is relabeled as its minimizer. Prediction-dependent optimization has an established literature (Sakaue and Oki, 2023, Sambharya, Hall, Amos, and Stellato, 2023); the additional object here is a computable original-law Bellman certificate that can license an action before another query and supplies a directly auditable training criterion.

7. TWO CONTROLS AND TWO CONTINUOUS INNOVATIONS

The absence of a state lattice does not remove action or innovation dimensions. We extend the query implementation to the previously specified two-control economy of Section 4: $a_1, a_2 \geq 0$, $a_1 + a_2 \leq c(x)$, the interaction cost is $a_1 a_2/4$, and the second innovation is an independent common-direction uniform shock. The original scalar model and its experiments are unchanged.

At a queried state, triangulate the feasible action simplex. Let v_0, v_1, v_2 be a triangle's vertices with verified lower endpoints l_0, l_1, l_2. If $\theta_j(a)$ are its barycentric coordinates, define

$$p(a) = \sum_{j=0}^{2} \theta_j(a) l_j - \frac{H}{2} \sum_{j=0}^{2} \theta_j(a) \|v_j - a\|_2^2. \tag{26}$$

This is the quadratic $H\|a\|_2^2/2 + b^\top a + c$ interpolating the lower endpoints. Its approximate minimizer is only a proposed support point. For any point $y \in \mathbb{R}^2$, including a numerically displaced point outside the triangle,

$$p(y) + \min_j \nabla p(y)^\top (v_j - y) \leq \min_{a \in \mathrm{co}(v_0, v_1, v_2)} p(a). \tag{27}$$

Thus an outward supporting-plane calculation, not an uncertified quadratic solver, supplies the lower endpoint.

THEOREM 11—Simplicial Bellman query: *For the two-control primitives,* (26) *is a lower minorant of the true Bellman objective. Taking the smallest valid triangle lower endpoint and the best feasible-action upper endpoint gives a lower–upper certificate for the full continuous simplex. Include the omitted capacity strip and actual action-rounding allowances. Returning when this gap is at most δ gives the same acquired-policy loss recursion as Theorem 7.*

A failed floating triangulation can be replaced by an exact-rational simplex enumeration. Under the stated positive precision slack, this recovery is finite at every finite horizon. With two innovations divided into q bins each, a valid local quadrature error is

$$\zeta = \frac{5\beta M_{t+1} d}{24576 q^2}. \tag{28}$$

If A_r bounds the number of action queries at a state with r dates remaining, a work account has the form

$$W_r \le C_d A_r + A_r q_r^2 W_{r-1}, \qquad C_d = O(d), \tag{29}$$

with separate charges for triangulation, arithmetic and output. There is no tensor state-table factor, but the descendant tree remains.

Proof. See Supplement, Section 3.4.

The executed algorithm uses longest-edge refinement of the lowest triangle and an outward terminal convexity-gap check. Its floating cap is followed by the stated recovery rule, not by an uncertified returned action. The implementation batches descendants but can allocate a descendant array before slicing it into query batches; a reported maximum query batch is therefore not a bound on total live state memory. We report process memory separately. The two-control execution is a verified extension of the NBO query backend, not evidence for a trained two-output selector.

8. FROZEN NEURAL CONTINUATIONS AND POLICY-COST PRECISION

The optimal-advantage estimator in Section 1 makes a policy comparison informative, but its conditional optimal-value queries are expensive. A fitted continuation can replace those queries without changing the estimand. The necessary certificate concerns its Bellman residual across states, not its distance from an unknown optimizer. This places the original NBO continuation, rather than only a proposed action, inside the evaluation calculation.

8.1. *Residual oscillation is the relevant range*

Consider the bounded expected-cost model with common terminal cost g, discounts $\beta_t \ge 0$, and a common initial-state law. Write $b_0 = 1$ and $b_t = \prod_{j<t} \beta_j$. Fix bounded measurable continuations $h_0, \dots, h_T$, conditionally on their construction. Define

$$Q_t^h(x,a) = c_t(x,a) + \beta_t P_t^a h_{t+1}(x),$$

$$r_t^h(x) = \inf_{a \in A_t(x)} Q_t^h(x,a) - h_t(x),$$

$$a_t^{h,\pi}(x) = Q_t^h(x, \pi_t(x)) - \inf_{a \in A_t(x)} Q_t^h(x,a).$$

The infimum is over the original continuous feasible set. Suppose verified constants give $0 \le a_t^{h,\pi} \le \eta_t^\pi$, and likewise for ν, throughout the true state domain. Let ω_t bound the oscillation of r_t^h, and ω_T that of $g - h_T$. Set

$$\Omega_h = \sum_{t<T} b_t \omega_t + b_T \omega_T, \qquad H_h^\pi = \sum_{t<T} b_t \eta_t^\pi.$$

These constants include acquisition, action rounding and arithmetic allowances wherever those operations affect the implemented policy or the verified functions. A small empirical residual alone is not such a certificate.

THEOREM 12—Frozen-continuation comparison: *Along the actual trajectory of π, put*

$$Z_h^\pi = \sum_{t<T} b_t \left[Q_t^h(X_t^\pi, \pi_t(X_t^\pi)) - h_t(X_t^\pi) \right] + b_T (g - h_T)(X_T^\pi). \tag{30}$$

For any coupling of the two original trajectory laws, $D_h = Z_h^\pi - Z_h^\nu$ satisfies

$$\mathbb{E} D_h = \mathbb{E}[J_0^\pi - J_0^\nu], \tag{31}$$

$$-\Omega_h - H_h^\nu \le D_h \le \Omega_h + H_h^\pi. \tag{32}$$

Moreover, at every initial state,

$$0 \le J_0^\pi - V_0^* \le \Omega_h + H_h^\pi. \tag{33}$$

Adding an arbitrary state-independent constant to each h_t changes neither D_h nor these oscillation bounds. With $h_t = V_t^*$, the result reduces to optimal-advantage centering.

The proof is in Supplement, Section 5.1.

The terminal correction in (30) is essential unless $h_T = g$. Conditional expectation, not pathwise equality with the realized cash-cost difference, supplies (31). The result uses the actual kernels and costs; it assumes neither a correct statistical transition

model estimated from data nor a convex neural fit. The pointwise policy certificate also applies conditionally after an adaptive choice among valid continuations. The independent-sample confidence statement below requires its additional sampling conditions.

For M conditionally fixed comparisons, suppose independent trajectory rows yield outward enclosures $[L_i, U_i]$ of D_h, intersected with its valid support. Write $R_h = 2\Omega_h + H_h^\pi + H_h^\nu$. A simultaneous interval with failure probability at most α is

$$\left[\overline{L} - R_h\sqrt{\frac{\log(2M/\alpha)}{2n}},\quad \overline{U} + R_h\sqrt{\frac{\log(2M/\alpha)}{2n}}\right]. \tag{34}$$

This follows by the bounded-sum inequality applied to both endpoint samples and a union bound. The numerical and sampling width allocations of Corollary 2 apply with R_h. Fitting, uniform residual qualification, conditional integration and any subsequent selection are distinct paid operations. In particular, a cache that adapts across rows changes the controller's history: its rows cannot be treated as independent evaluations of one frozen Markov policy. A fixed cache or an independent episode-reset design is sufficient to retain the stated sampling construction.

8.2. *Conditional integration of an actual neural continuation*

A shallow rational affine–ReLU continuation has the form

$$h(y) = c + \ell^\top y + \sum_{j=1}^{K} v_j (w_j^\top y + u_j)_+.$$

Its readout weights may have either sign. In the original investment laws the next state, conditional on (x, a), is affine in one or two independent centered uniform innovations. A neuron therefore has projected center z and nonnegative projection radii A, B. Write $z_+ = \max\{0, z\}$.

PROPOSITION 13—Exact conditional ReLU moments: *For independent uniform projections of radii $A, B > 0$, the conditional means of a neuron are*

$$\mathcal{M}_1(z; A) = \frac{(z + A)_+^2 - (z - A)_+^2}{4A}, \tag{35}$$

$$\mathcal{M}_2(z;A,B)=\frac{1}{24AB}\big[(z+A+B)_+^3-(z+A-B)_+^3 - (z-A+B)_+^3+(z-A-B)_+^3\big]. \tag{36}$$

A zero projection radius is removed; if all radii vanish the mean is z_+. Consequently $P^a h$ is a finite exact rational calculation for rational primitives, states, actions and weights. For n trajectories and at most K neurons per date, its arithmetic-operation count is $O(nTKd)$, exclusive of fitting, trajectory construction and uniform qualification. No optimal-value descendant query occurs in this calculation.

The proof is in Supplement, Section 5.2.

The operation count does not bound rational bit complexity. Near-zero projected radii can also make a naive floating evaluation unstable; directed rounding or rational evaluation, including the zero-radius branches, is required. The executed moment catalogue checks the rational identities against an independently written polygon integral. It is a conditional-integration experiment, not a new trained-policy cost comparison. The formulas do not imply that the achieved residual oscillation of an arbitrary fitted network is small.

8.3. *Qualifying a fitted continuation without a Bellman tree*

Uniform qualification need not query V^*. Suppose Q_t^h is Lipschitz in state and action with constants $L_{x,t}, L_{a,t}$, and h_t has constant $L_{h,t}$, all in ℓ^1. In the investment capacity simplex, an action feasible at y can be repaired continuously at x with displacement at most $\kappa\|x-y\|_1$, where $\kappa = 1/(8d)$. Thus a valid residual Lipschitz constant is

$$L_{r,t} = L_{x,t} + \kappa L_{a,t} + L_{h,t}.$$

Let a finite state cover have fill radius ρ, and let $[l_j, u_j]$ enclose r_t^h at every cover point. Then

$$\min_j l_j - L_{r,t}\rho \le r_t^h(x) \le \max_j u_j + L_{r,t}\rho \tag{37}$$

throughout the cube. At a cover point, a feasible action mesh of fill radius s gives a lower infimum bound by subtracting $L_{a,t}s$ from the smallest action-value lower

endpoint; the smallest upper endpoint is an upper bound. Exact moments can supply those action endpoints. None of these steps requires convexity of h or a true-value curvature minorant for Q^h.

For N_t state-cover points and A_t actions per point, this qualification costs $O(\sum_t N_t A_t K_{t+1} d)$ arithmetic operations, plus certification of the moduli and any actor-specific greedy-gap bound. Its cover can still be expensive in high dimension. The result replaces a recursive optimal-value oracle by an explicit approximation-and-qualification tradeoff; it does not make that qualification free. The residual-cover proof, the action-mesh argument and the terminal allowance are given in the supplement. Value-function control variates have an established variance-reduction literature (Munos, 2006); the additional requirement here is an original-law, whole-domain certificate for the fitted NBO continuation and the actual acquired policy.

9. FITTING A CONTINUATION FOR QUALIFICATION

A continuation used to evaluate a policy should be fitted to the error that determines the evaluation cost. The residual-cover bound in (37) supplies such an objective. We now make its optimization constructive. The future continuation is held fixed at each backward step; neither an optimal-value oracle nor success of unrestricted hidden-layer training is assumed.

Fix a date and suppress its subscript. Let $x_1, \dots, x_N$ be a state cover of radius ρ in the ℓ^1 metric, and suppose

$$y_j^- \le (\mathcal{T} h_+)(x_j) \le y_j^+.$$

These are verified bounds on the original continuous-action infimum, not only on a proposal menu. Write $h_\theta = \sum_{k=1}^K \theta_k \phi_k$, where the fixed features have verified Lipschitz constants L_k and $|\theta_k| \le B_k$. The constant feature is allowed with $L_k = 0$. Let C be a verified Lipschitz constant of $\mathcal{T} h_+$, including state-dependent feasibility repair. With $F_{jk} = \phi_k(x_j)$, define

$$\mathcal{W}(\theta) = \max_j (y_j^+ - F_j\theta) - \min_j (y_j^- - F_j\theta) + 2\rho \left(C + \sum_k L_k |\theta_k| \right). \tag{38}$$

THEOREM 14—Qualification-optimal readout: *For every feasible* θ, $\operatorname{osc}(\mathcal{T}h_+ - h_\theta) \le \mathcal{W}(\theta)$. *Minimizing* (38) *is a finite linear program with* $2K+2$ *variables and* $2N+2K$ *inequality constraints, apart from variable bounds. For probability vectors* p, q *on the nodes, set* $\lambda_k = 2\rho L_k$ *and*

$$\mathcal{D}(p,q) = p^\top y^+ - q^\top y^- + 2\rho C - \sum_k B_k \big(|[F^\top (q-p)]_k| - \lambda_k\big)_+. \tag{39}$$

Then $\mathcal{D}(p,q) \le \inf_\theta \mathcal{W}(\theta)$, *with equality for some* p, q. *Thus any feasible readout and exact probability vectors certify an achieved optimization gap of at most* $\mathcal{W}(\theta) - \mathcal{D}(p,q)$.

The proof is in the Technical Supplement. It constructs the primal constraints and dual support function explicitly. Rationalizing a solver's proposed readout and probability vectors gives a checkable certificate; a convergence flag is not the certificate. Label intervals, features, moduli, coefficient bounds and the cover remain part of the numerical contract.

Applying the theorem backwards gives a fitted continuation at every date. Its achieved ranges $\mathcal{W}_t$, together with the terminal oscillation and the actual implemented greedy-gap bounds, can be substituted directly into Theorem 12. This connects the attained fitting objective to both original-optimum regret and the range of the original-law cost observation. Each readout minimizes its stated one-step qualification bound conditional on the fixed future. It does not minimize the joint multi-date range, the true policy loss, or a hidden-feature objective. The next backward step recomputes its labels and moduli from the continuation actually returned.

The label construction in (37) uses exact conditional moments for a shallow ReLU future and a verified continuous-action cover. Its work is charged in addition to the linear program, coefficient rationalization, modulus calculation and dual check. A conventional basis with the same number of coefficients receives the same labels and the same objective; it uses its own valid feature moduli. This is a comparison of qualified approximations, not an assertion that different bases span the same functions or that a neural encoding has an intrinsic advantage.

10. SEQUENTIAL QUALIFICATION OF POLICY COSTS

A fixed-sample interval does not by itself justify repeated inspection until a controller appears acceptable. Nor do cost observations from changing controllers identify the last controller's cost. We extend the frozen-continuation identity to predictable episodes and distinguish these targets. This uses established time-uniform concentration principles (Howard, Ramdas, McAuliffe, and Sekhon, 2020); confidence sequences have also been used for gated deployment in contextual bandits (Karampatziakis, Mineiro, and Ramdas, 2021). Here the observations are verified, continuous-law NBO cost contrasts, not importance-weighted bandit rewards.

Let $\mathcal{F}_{i-1}$ contain all information before episode i. Conditional on it, choose two complete controllers, a common initial law, a continuation h_i fixed through the episode, and valid coefficients of Theorem 12. The resulting cost contrast D_i has conditional mean Δ_i equal to the original-law expected cost difference and predictable support $[a_i, b_i]$. Let $[L_i, U_i] \subseteq [a_i, b_i]$ enclose D_i for the entire continuous innovation cell. Write $V_n = \sum_{i \le n} (b_i - a_i)^2$. For fixed positive numbers λ_j, w_j, $1 \le j \le J$, with $\sum_j w_j \le 1$, define

$$B_n = \min_{1 \le j \le J} \left\{ \frac{\log(2/(\alpha w_j))}{\lambda_j} + \frac{\lambda_j V_n}{8} \right\}. \tag{40}$$

THEOREM 15—Predictable original-law cost certificate: *Under the preceding conditional-law, support and enclosure conditions, with probability at least* $1 - \alpha$, *simultaneously for every* $n \ge 1$,

$$\frac{\sum_{i \le n} L_i - B_n}{n} \le \frac{1}{n} \sum_{i \le n} \Delta_i \le \frac{\sum_{i \le n} U_i + B_n}{n}. \tag{41}$$

Each endpoint may also be intersected with the corresponding average predictable support endpoint. The assertion holds at every almost surely finite stopping time. If the complete controller pair and initial-law/reset contract stay fixed, then $\Delta_i = \Delta$, *even when the qualified continuation changes predictably between episodes. Otherwise the target is the average conditional cost difference, not the last controller's cost.*

The proof applies conditional Hoeffding bounds to each endpoint and Ville's inequality to a finite collection of nonnegative supermartingales. It requires no independent or identically distributed deployment contexts. It does require the stated conditional transition laws. Within-episode cache adaptation is allowed only when the controllers are treated as history-dependent algorithms and the greedy-gap qualification covers every memory state that can occur. Repeating the same program identifier while carrying a different cache state does not establish a fixed conditional target.

The implementation uses a finite dyadic collection of λ_j and exact rational upper bounds for the logarithms. These give conservative, computable versions of (40). It reserves the support and target identity before accepting an episode. A failed numerical evaluation consumes the full previously qualified support interval and its actual work; it is not deleted after inspecting its outcome. A failure of the support certificate itself stops qualification.

COROLLARY 16—Costed replacement after optional stopping: *For a fixed target pair, let $[l_N, u_N]$ be the interval at a finite stopping time N. Let a declared deployment count be $m \geq 1$, a one-time replacement and qualification charge be $F \geq 0$, and the incremental per-deployment charge be c, all in the same units as the policy cost. On the simultaneous event of Theorem 15, the net incremental cost lies in*

$$[l_N + F/m + c,\ u_N + F/m + c].$$

Consequently $u_N + F/m + c < 0$ certifies strict net improvement. Inclusion of $[l_N, u_N]$ in a declared equivalence band certifies equivalence at that band, not strict superiority. New adaptively selected controller pairs may be qualified in separate epochs with conditional error allocations whose sum is at most the overall error budget.

A wall-clock reading is not a monetary charge. Measured setup, validation, rejected evaluations and successful retrievals are retained separately; prices are explicit decision inputs. The number of qualification episodes is not automatically the number of deployments over which a setup cost is amortized. These distinctions let the original investment decision use a valid stopping rule without silently changing either its estimand or its units.

11. TRANSPORTING PURCHASED BELLMAN WITNESSES

A second way to control the descendant cost is to reuse a calculation already purchased. Exact memoization only recognizes the same state. The following rule transfers a lower Bellman bound and its feasible upper witness to a nearby state, after repairing capacity and accounting for the full displacement. It is compatible with either a neural or a conventional generator of the original certified record.

11.1. *An original-optimum transport certificate*

At remaining horizon r, a record at y contains an action $a_y \in A(y)$ and numbers satisfying

$$L_y \le V_r^*(y) \le Q_r^*(y, a_y) \le U_y.$$

It also identifies the economic primitives, remaining horizon and numerical contract under which the record was obtained. A file hash detects changed bytes; validity of the inequalities is a separate premise. For the original scalar and two-control economies, let

$$L_V = G_r, \qquad L_Q = \frac{8}{d} + \frac{11\beta}{16} G_{r-1}, \qquad L_A = \frac{13}{16} + \frac{3\beta d}{8} G_{r-1}.$$

These bound state variation of V_r^* and state and action variation of Q_r^*. The action comparison may be made on the ambient simplex $a \ge 0$, $\sum_i a_i \le 1/4$: the original transition remains in the state cube there, even if a particular action violates the smaller capacity at one endpoint.

At a new state x, repair a_y to an actual lattice action $\widetilde{a} \in A(x)$. Set $D = \|x - y\|_1$ and $\Delta_a = \|\widetilde{a} - a_y\|_1$. Define

$$L(x) = \max\{0, L_y - L_V D\}, \qquad U(x) = U_y + L_Q D + L_A \Delta_a. \tag{42}$$

The truncation at zero uses the nonnegative costs of this economy. It is not part of a general signed-cost transfer rule.

THEOREM 17—Reuse after capacity repair: *Under the stated record and modulus conditions,*

$$L(x) \le V_r^*(x) \le Q_r^*(x,\widetilde{a}) \le U(x).$$

Hence outward endpoints with $\overline{U}(x) - \underline{L}(x) \le \delta$ permit the actual repaired action to be returned without a new Bellman query. An identity mismatch, failed feasibility check or excessive width requires a new verified calculation; proximity alone licenses no reuse.

The same assertion holds after any history-dependent selection of valid records. At each future date, suppose the selected record or fresh query supplies the original local-gap and acquisition premises of Theorem 7. The resulting, possibly history-dependent controller retains that theorem's original-optimum loss sum. It need not coincide with the action returned by a fresh search at the same tolerance.

The proof is in Supplement, Section 5.3.

The implemented repair first rounds the new capacity downward on the 2^{-32} lattice, scales a vector action proportionally when necessary, and rounds each coordinate downward. The certificate charges its exact resulting ℓ^1 displacement, not merely the change in capacity. Retrieval may use approximate distances or a learned index as an ordering heuristic. Every returned hit must still satisfy (42) under outward arithmetic.

11.2. *A finite-cover account and its premises*

A small number of hits does not establish a uniform reuse complexity bound. Such a bound requires a fill precision, a coverage argument and a retention rule. Let the action quantum be q, let $m \in \{1,2\}$, and suppose every fresh record is purchased with gap at most $\gamma\delta$, where $0 < \gamma < 1$. The repair just described obeys

$$\Delta_a \le \kappa D + (m+1)q, \qquad \kappa = 1/(8d).$$

Consider the exact-endpoint transport rule, or an outward rule with a known total additional width allowance $\xi \ge 0$. Define

$$\rho_* = \frac{(1-\gamma)\delta - L_A(m+1)q - \xi}{L_V + L_Q + \kappa L_A}. \tag{43}$$

PROPOSITION 18—Paid fills and complete reuse work: *Suppose $\rho_* > 0$, all valid records are retained, the context is fixed, and every stored candidate is checked before a miss is declared. A record within distance ρ_* certifies a hit. On the unit state cube, the number M_r of fresh records at remaining horizon r is at most*

$$M_r \le \left(1 + \frac{d}{\rho_*}\right)^d. \tag{44}$$

If $C_r(\gamma\delta)$ bounds the complete cost of a fresh fill, a workload of B requests obeys

$$W_B \le W_{\text{setup}} + M_r C_r(\gamma\delta) + \sum_{b=1}^{B} \left(W_b^{\text{lookup}} + W_b^{\text{check}} + W_b^{\text{load/store}}\right). \tag{45}$$

The lookup and check costs include rejected candidates. Naive retrieval uses $O(M_r d)$ arithmetic operations per request, apart from rational bit costs. Storage, eviction, invalidation and context changes require their own additional fill accounts when the stated retention premises do not hold.

The proof is in Supplement, Section 5.4.

This bound makes the dimensional price of uniform reuse explicit. It is not a dimension-free caching theorem. It also separates a first expensive Bellman fill from later transport checks. Caching at the workload root can avoid all descendants on a hit, but does not accelerate the descendants of a miss. The executed bounded cache uses 128 records and the requested fill tolerance, not the tighter $\gamma\delta$ premise. Its operation counts below are direct observations, not an application of (44). A cache at recursive descendants would need a separate execution and ledger.

12. CONTINUATION EVALUATION AND PAID QUERY REUSE

The evidence separates three questions: whether a certified readout returns an admissible policy, whether its actual expected cost meets a declared margin, and whether a purchased calculation can be reused profitably. The preceding training and inference catalogue remains unchanged. The new execution fixes localized query workloads and exact conditional-moment checks before running them. It changes neither the economic primitives nor the continuous feasible-action target.

12.1. *Retained trained policies and economic precision*

The retained 188-process catalogue is reproduced in Appendix, Section F. Hat–ReLU and Bernstein readouts purchase the same teacher information and have exactly $5d$ coefficients; the achieved objective and box dual gaps are recomputed rationally. Independent validation and sixty-four-block reuse receive separate work accounts. Both their favorable and unfavorable repetitions remain visible. In particular, the eight-state task does not identify a learned complete-work winner.

The direct policy-cost experiment fixes its sample count and numerical precision before execution. All twelve original-law paired intervals lie within the declared equivalence region $[-1/256, 1/256]$, and all contain zero. The two adaptive-minus-hat–ReLU intervals are approximately $[-0.00068263, 0.00070764]$ and $[-0.00160291, 0.00160253]$. Their shared inference services cost about 9.03 and 408.63 seconds. These are informative quality comparisons with expensive qualification, not an identified neural cost ranking or a free inference improvement. The complete precision and adoption tables are retained in the appendix.

The frozen-continuation theorem supplies a different way to evaluate that expected-cost estimand. Its moment component is executed here; a new trained critic meeting the old equivalence margin at lower complete qualification cost is not presumed. The earlier optimal-advantage observations are not counted again as samples of the new formula.

12.2. *A fixed original-law query catalogue*

The new design has five tasks: two scalar-control tasks at $(d, T) = (2, 2), (8, 3)$ and three two-control tasks at $(d, T) = (2, 2), (2, 3), (2, 4)$. Each task has three rules: fresh calculation, exact-key reuse and geometric witness transport. The source commitment fixes the task list, tolerances, repetition counts, 128-record capacity, invalidation schedule, workload order and 900-second per-process limit. There are 36 prescribed complete processes. None trains a new action predictor or loads a precomputed conventional policy.

A workload starts at alternating coordinates $0.4, 0.6$. Odd requests repeat that state. Predetermined even requests perturb one coordinate by 2^{-20}. Except in the

four-date stress case, a late distant request has every coordinate equal to 0.75, and a specified earlier block invalidates the context. The first four tasks have 32, 16, 24 and 8 requests; the four-date task has four. The first three tasks have three timing repetitions, the three-date vector task two, and the four-date task one. Rules are executed in a fixed cyclic order across repetitions.

Every miss calls the unchanged original recursive verifier at the declared tolerance. Every hit uses the exact state-transfer and capacity-repair test. The cache is serialized, hashed, reloaded and validated after every block. The receipt begins before process creation and ends after exit, including imports, all descendant work, file output and shutdown; writing the parent receipt itself is separate publication overhead. A single CPU is pinned. Available start and end frequency information is recorded, but frequency is not locked. Repetitions support descriptive comparisons, not a hardware-uniform or population timing theorem.

The table counts Bellman action evaluations at all descendants, not only workload roots. Geometric hits at a nonidentical state distinguish transport from exact memoization. Their capacity-repair count is separately recorded, including downward repairs that change the returned action. The distant request and context invalidation test whether a cheaper rule refuses an unjustified reuse rather than forcing a hit. An unchanged numerical law with a new context identifier still triggers the declared conservative invalidation.

The four-date task retains the first expensive tree. A hit avoids that tree on a later request, whereas a miss pays for it again. This is a local repeated-query improvement under the original optimum contract, not a disappearance of the long-horizon first-fill cost. The trace records every source witness, state displacement, repaired action and outward endpoint. The absence of a state lattice does not justify omitting either the descendant count or resident memory.

The full repetition and storage ledger is in Appendix, Section M. Prefix clocks there begin after module loading and are labeled as body clocks; they are not substituted for complete process receipts. The bounded cache and requested fill tolerance do not satisfy the stronger retained-cover premises of Proposition 18, so that proposition’s uniform packing count is not presented as an empirical result.

TABLE I

COMPLETE ORIGINAL-LAW QUERY WORKLOADS WITH AND WITHOUT REUSE

Task	Rule	Fills	Hits	Bellman queries	Seconds
S2/2	Fresh	32	0	490	1.166
S2/2	Exact	7	25	115	0.765
S2/2	Geometric	3	29	55	0.669
S8/3	Fresh	16	0	1,610	2.740
S8/3	Exact	9	7	959	1.867
S8/3	Geometric	3	13	401	1.167
V2/2	Fresh	24	0	397	2.471
V2/2	Exact	7	17	125	1.216
V2/2	Geometric	3	21	61	0.916
V2/3	Fresh	8	0	25,534	22.047
V2/3	Exact	5	3	17,023	14.380
V2/3	Geometric	3	5	11,349	9.591
V2/4	Fresh	4	0	5,085,947	347.972
V2/4	Exact	2	2	2,542,973	168.527
V2/4	Geometric	1	3	1,271,487	84.069

S and V denote scalar and two-control tasks; numbers are state dimension and horizon. Queries include every recursive descendant. Seconds are medians of complete processes: three repetitions for S2/2, S8/3 and V2/2, two for V2/3, and one stress observation for V2/4. All prescribed records are retained. These are localized workload comparisons, not population policy-cost samples.

12.3. *Independent conditional-moment and transport checks*

The rational integration catalogue uses four (d, m) specifications, namely $(2,1), (8,1), (2,2), (8,2)$, and 64 state–action contexts per specification. Each context has sixteen rational neurons. The original bilinear state drift, action exposure and uniform innovation projections are evaluated exactly. An independent checker clips the corresponding innovation rectangle and integrates the affine neuron through polygon area and first moments. It imports no production moment, cache or interval-query routine.

TABLE II

FOUR-DATE, TWO-CONTROL FIRST-FILL AND COMPLETE WORKLOAD ACCOUNT

Rule	First-fill queries	All queries	Seconds	RSS (KiB)
Fresh	1,271,487	5,085,947	347.972	99,212
Exact	1,271,487	2,542,973	168.527	99,216
Geometric	1,271,487	1,271,487	84.069	99,212

Four requested root states, two independent innovations, and common local tolerance 512/14911. The first-fill column counts its full descendant tree. Exact reuse recognizes identical states; geometric reuse additionally transfers a valid witness to a nearby state. RSS is total process peak memory, not logical cache bytes. One repetition is descriptive only.

TABLE III

INDEPENDENT CHECKS OF THE NEW OPERATIONS

Quantity	Count
Prescribed complete processes	36
Returned / failed	36 / 0
Checked root returns	708
Checked reuse returns	357
Different-state reuse returns	127
Repaired-action reuse returns	18
Independent exact neuron integrals	4096
New independent policy-cost rows	0

The checker imports no production numerical kernel. It verifies each new transfer against a prior paid fill and independently integrates each rational neuron on the original innovation support. It does not independently regenerate the internal optimal-value trees of inherited fills.

The same independent checker verifies every new hit from its earlier paid fill, including feasibility, action lattice, context, original primitive moduli, exact transport endpoints and outward conversion. This is independent reconstruction of the new transfer operation. The inherited recursive fill endpoints remain a disclosed dependency; their entire internal trees are not independently regenerated by this checker. The preceding independent full-bin cash-path calculations and the histor-

ical accumulation-aliasing counterexample remain in the retained appendix and companions.

The moment tests establish exact conditional integration, not the accuracy of an unobserved fitted critic. The query workload establishes a localized certificate-preserving reuse comparison, not independent expected policy-cost observations. A history-dependent transported policy is not identified with the frozen policies in the old cost-inference study. Its economic adoption therefore requires its own eligible cost account or a decision maker's declared worst-case accuracy requirement, together with the measured fill, reuse and qualification charges.

12.4. *Constructed continuations and attained qualification*

The readout theorem is executed in four predeclared backward constructions in the original two-state, two-date, scalar-control economy. Two midpoint covers, with eight or sixteen cells per coordinate, are crossed with additive shallow ReLU and degree-four Bernstein features. Each basis has nine coefficients bounded by eight. At each date the future is the continuation actually constructed at the next date. Exact conditional moments supply thirty-three continuous feasible action values per node and thirty-three robust, downward-quantized acquired-action proposals. The original state-dependent capacity and continuous innovation law are unchanged.

TABLE IV

ATTAINED CONTINUATION QUALIFICATION IN THE ORIGINAL INVESTMENT ECONOMY

n	Basis	Ω_h upper	Regret upper	Max. dual gap
8	ReLU	5.39764	8.46716	0.000000014
8	Bernstein	5.36565	8.11941	0.000000000
16	ReLU	3.73299	5.28939	0.000000014
16	Bernstein	4.43474	5.81890	0.000000000

Each row is one predeclared, complete two-date construction. Bounds are rounded upward; the exact fractions and dates are retained. Regret upper denotes the proved original-optimum account, not an estimated loss or an attained economic equivalence margin. The maximum dual gap concerns only the declared fixed-feature, fixed-future readout objectives.

The table reports the attained two-date residual account Ω_h, the original-optimum bound $\Omega_h + H_h^\pi$, and the largest exact one-step optimization gap. A smaller qualification objective is not itself an identified expected-cost gain. The matched least-squares ablation in the appendix uses each service's identical features and fixed-future labels; it tests the fitting objective, not a separately propagated controller. Both fitting and uniform qualification are paid, and the explicit state cover is tensor based.

All eight datewise optimization gaps are below 1.4×10^{-8}, against the predeclared tolerance $1/65536$. Their qualified residual bounds are strictly below the matched label-loss bounds; the reductions range from about 18 to 63 percent. The resulting global regret bounds remain conservative, as the table makes explicit. Those objective reductions do not establish a narrow economic equivalence interval or cheaper qualification at the earlier $1/256$ margin.

Every new action value is independently reconstructed using the retained original rational terminal formula or a different exact integration method for the fitted continuation. Continuous-action lower covers, whole-cell choices and primal–dual bounds are then recomputed. These constructions add actual qualified continuations, not new independent cost observations. The stopping theorem has separate exact finite-model regressions; the earlier fixed-sample equivalence intervals are not retroactively given an anytime interpretation.

13. CONCLUSION

Neural Bellman Operators connect the construction of a continuation to feasible decisions under the original economic law. The witness-producing implementation has an explicit own-future error account. Recursive action verification permits trained proposals without allowing fitting error to alter the policy certificate. The present additions place a frozen neural continuation directly in original-law policy-cost evaluation and permit purchased original-value witnesses to be reused after a verified capacity repair.

The resulting resource distinctions are substantive. Exact conditional moments remove optimal-value descendants from the evaluation of a qualified shallow continuation, but fitting and uniform residual qualification remain paid operations.

Transported witnesses remove descendants on a certified cache hit, but the first fill and every later miss retain their original horizon cost. A history-dependent reuse controller retains a loss guarantee; it is not automatically the same policy used in an earlier independent cost experiment.

The retained cost-inference evidence identifies equivalence at a declared numerical margin, not strict superiority. The new query experiment tests localized reuse, not universal neural speed or calibrated welfare. Together with the constructive theorems, these results identify what can be certified, reused and economically compared, and what resources each step requires. They extend the original NBO policy-evaluation and improvement objective while preserving every historical application, assumption and unfavorable comparison for renewed scrutiny. The qualification-optimal readout gives a direct link from an attained fitting objective to a uniform NBO cost and regret certificate. Its exact dual distinguishes verified optimization accuracy from a solver's status. Predictable original-law cost intervals allow qualification to stop when a declared economic criterion is met, while keeping fixed-policy and adaptive-deployment targets separate. Both additions use the original investment primitives and pay for approximation, uniform qualification and numerical evaluation; neither infers a population ranking from a better training objective.

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